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Bayesian Model Selection for Raman Spectroscopy of *E. coli*: Determining the True Information Content of Spectroscopic Data

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Declaration

I hereby certify that this project report, which is approximately five thousand words in length, has been written by me at the School of Physics and Astronomy, Queen Mary University of London, that all material in this dissertation which is not my own work has been properly acknowledged, and that it has not been submitted in any previous application for a degree.

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Abstract

The analysis of Raman spectroscopic data is limited by noise and strong overlap between spectral contributions from different biomolecules. In this project, Raman spectra of the bacterium *Escherichia coli* in two different growth phases (log and stationary) were modelled as linear combinations of 15 individually obtained reference spectra (of biomolecules typically contained in *E. coli*). Bayesian model selection using the Bayes Factor Integral (BFI) was then applied to determine the optimal model complexity and hence the true information content of the spectra. Unlike other commonly used figures of merit such as AIC and BIC, the BFI accounts for parameter correlations, which is particularly important for heavily overlapped data. The analysis identified optimal models containing 8 biomolecular components for the log phase and 10 for the stationary phase, whilst conventional figures of merit generally favoured larger model sizes. The BFI therefore offers an objective stopping rule for determining how many biomolecular components the data genuinely supports, and avoids unjustified additions to the model.

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1 Introduction

The analysis of spectroscopic data has been an area of interest for many years, with many possible approaches to extracting meaningful insights. Raman microspectroscopy in particular is quite challenging to interpret due to signal overlap, with most biological research relying on assigning individual peaks to specific biomolecules, or on simplifying the data using techniques like principal component analysis (PCA). While these approaches have their merits, the information they provide is largely qualitative [Gong et al., 2025].

Raman scattering was first observed by Raman and Krishnan [1928], leading to Raman microspectroscopy decades later. This is an imaging technique based on the exchange of energy between light and matter, capable of detecting and quantifying biomolecules within single cells without the need for labelling or cell disruption [Butler et al., 2016, Lee et al., 2021]. When a laser is directed at a cell, a fraction of photons exchange energy with its vibrational modes, referring to the oscillations of atoms about their equilibrium positions. The difference in energy between the incident and scattered photon is called the *Raman Shift*. Since the vibrational modes within the molecules of a cell are determined by its chemical bonds and atomic structure, the pattern of Raman shifts acts as a unique biomolecular fingerprint [Butler et al., 2016]. This makes it especially useful for the analysis of bacterial systems, where preserving the environment of the cell is important. However, a single cell contains many molecules with similar chemical bonds, which produces strong overlap in the resulting Raman spectrum. Combined with experimental noise, this overlap makes it difficult to determine which components contribute genuinely to the measured signal [Gong et al., 2025].

This project focuses on datasets from the bacterium *E. coli* in two distinct physiological states during growth: exponential (log) phase, and stationary phase [Wolfe, 2005]. The cellular composition changes between these phases, and being able to detect and compare these differences is important for studying variation between individual bacteria within a population [Gong et al., 2025]. Figure 1.1 shows the log-phase spectrum analysed in this project. Distinguishing genuine signal from noise is the main challenge of the analysis.

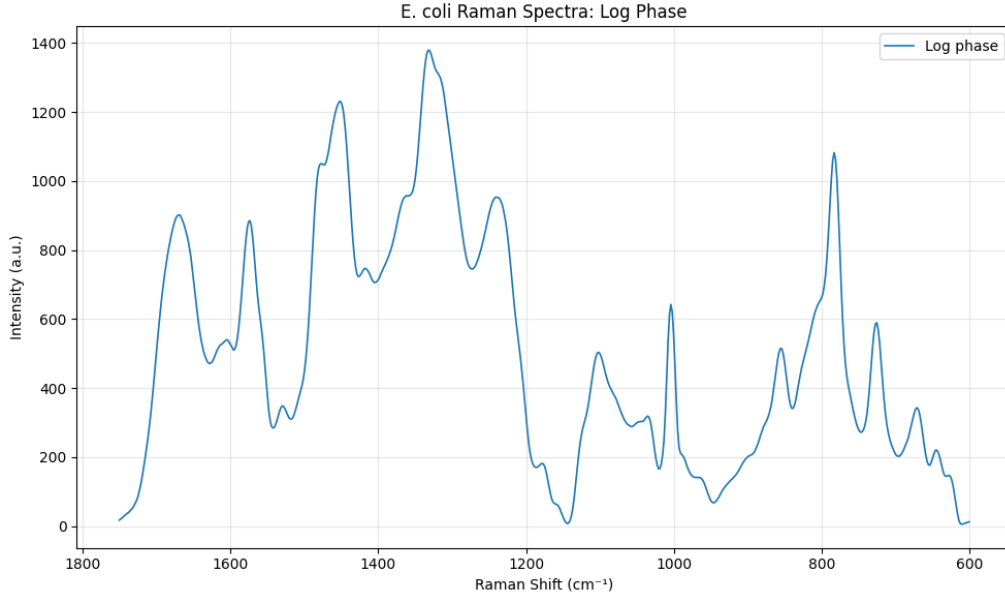


Figure 1.1: Raman spectrum of *E. coli* in exponential growth phase. The spectrum contains overlapping contributions from multiple biomolecules, making it difficult to identify individual components.

Interestingly, [Gong et al. \[2025\]](#) developed a different approach to this problem. Instead of labelling peaks or reducing dimensionality, they fitted the cellular Raman spectra as linear combinations of reference spectra obtained from individual biomolecules. Their method focused on conducting analysis after fitting the model, but the same linear combination framework can be built up in steps rather than fitting it all at once. With this, depending on how reference components are selected, one would be able to interpret contributions more efficiently. This approach does however raise a question: how many reference components should be included in the fit, and how should the stopping point be determined? One may look at the Nyquist-Shannon sampling theorem, which sets a theoretical maximum on the number of parameters a dataset can support, but this is only strictly true for the case of independent parameters. In reality, the actual usable information content of spectroscopic data falls well below this maximum, because parameters are correlated and noise further limits how many components can be reliably extracted [[Haddad et al., 2024](#)].

Standard figures of merit such as χ^2 , the reduced χ^2_ν , the Akaike information criterion (AIC), and the Bayesian information criterion (BIC) each attempt to balance fit quality against model complexity, but none of these stopping criteria properly account for parameter correlations [[Haddad et al., 2024](#)]. Recently, [Haddad et al. \[2024\]](#) applied the Bayes Factor Integral (BFI) as a figure of merit for model com-

parison in the analysis of extended X-ray absorption fine structure (EXAFS) data. The BFI stems from Bayesian statistics and is able to penalise unnecessary model complexity and parameter correlations [Dunstan et al., 2022]. It was shown to be superior to conventional figures of merit in selecting the correct structural model for both reference and unknown materials [Haddad et al., 2024].

In this project, we apply the BFI approach, originally developed for EXAFS analysis, to the Raman spectroscopy of *E. coli*. We determine its true information content by using the reference library, paired with BFI-guided model selection to identify the optimal number of biomolecular components needed to describe the data. This is applied to spectra from cells in both exponential and stationary growth phases, allowing for comparison across different physiological states. The performance of the BFI is then compared to conventional figures of merit to demonstrate the importance of accounting for parameter correlations when determining optimal model complexity.

2 Model Selection Framework

2.1 Linear Combination Fitting

Linear combination fitting models the cellular spectrum as a linear combination of reference spectra measured from known biomolecules. The reference spectra are Raman spectra of individually purified biomolecules, each measured separately under the same conditions as the cellular spectra [Gong et al., 2025]. These reference spectra can be used as building blocks to reconstruct the more complex spectrum of *E. coli*. This is possible because a molecule’s Raman spectrum is determined by its bond structure, the same biomolecule produces a similar spectral pattern whether measured individually or within a cell. And so, the cellular spectrum is modelled as a linear superposition of the biomolecules in the reference spectra, and the fitting process determines their relative contributions.

If the spectrum is represented as a vector $\mathbf{y}$ of intensity values at n wavenumber points, the model is written as:

$$y(\nu) = B(\nu) + \sum_{i=1}^N c_i R_i(\nu) + \epsilon(\nu) \quad (2.1)$$

where $B(\nu)$ is a baseline term accounting for residual background, $R_i(\nu)$ are the reference spectra of N biomolecular components, and $\epsilon(\nu)$ is the noise term.

In matrix form, this becomes:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon} \quad (2.2)$$

where $\mathbf{A}$ is a matrix containing the baseline terms together with the selected reference spectra, and $\mathbf{x}$ is the vector of coefficients to be determined by the fit. The construction of matrix $\mathbf{A}$ is described in Section 3.2. The matrix form makes it easier to solve using least-squares methods in Python, as it is essentially a linear regression problem, with the columns in $\mathbf{A}$ as the predictors.

The fitting is performed using bounded linear least squares, minimising:

$$\|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2 \quad (2.3)$$

with the constraint $c_i \geq 0$, to prevent the model from assigning negative contributions to biomolecules (as they are not physically possible).

2.2 The Overfitting Problem

The challenge with linear combination fitting is determining the appropriate number of biomolecular components N to include. Including too many reference spectra would lead to overfitting, and while an overfitted model may have a more accurate fit, it assigns meaningless contributions to components that the data does not actually justify. A stopping criterion is needed here to determine which reference spectra actually contribute.

Several standard figures of merit attempt to balance fit quality against model complexity:

- **Chi-squared:** $\chi^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$, this is the sum of squared residuals between the measured values y_i and the model predictions $\hat{y}_i$. Under the constant-noise assumption used in this project (see Section 2.3), this is proportional to the usual chi-squared statistic. It always improves when parameters are added, and thus provides no guidance on when to stop.
- **Reduced chi-squared:** $\chi_\nu^2 = \frac{1}{n-k} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, which normalises by degrees of freedom.
- **R^2 :** $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$, this is the fraction of variance explained, like χ^2 , it improves with added parameters and has no penalty for complexity.
- **AIC:** $\text{AIC} = n \ln\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right) + 2k$, based on the likelihood with a linear penalty of 2 per parameter [Akaike, 1974].
- **BIC:** $\text{BIC} = n \ln\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right) + k \ln n$, similar to AIC but with a stronger penalty that scales with the number of data points [Schwarz, 1978].

While AIC and BIC do penalise model complexity, none of the above criteria account for *correlations* between fitting parameters [Haddad et al., 2024]. In spectroscopic fitting, any reference spectra with similar spectral shapes will overlap in the spectrum. Their coefficients are therefore correlated. Ignoring these correlations

can lead to the selection of incorrect models, as demonstrated by [Haddad et al. \[2024\]](#) in the context of EXAFS analysis.

2.3 The Bayes Factor Integral

Given observed data D and a model M_i , Bayes' theorem gives the posterior probability of the model [\[Gregory, 2005\]](#):

$$P(M_i | D) = \frac{P(D | M_i) P(M_i)}{P(D)} \quad (2.4)$$

where $P(D | M_i)$ is the marginal likelihood (or evidence) for the model, $P(M_i)$ is its prior probability, and $P(D)$ is a normalisation constant. The evidence measures how well the model as a whole explains the observed data. When comparing two models, the ratio of their evidences, i.e. the Bayes factor, quantifies which model is better supported by the data [\[Dunstan et al., 2022, Gregory, 2005\]](#).

The Bayes Factor Integral (BFI), developed by [Dunstan et al. \[2022\]](#) and applied by [Haddad et al. \[2024\]](#) for EXAFS analysis, provides an expression for the evidence of a model with k fitting parameters:

$$\text{BFI} = (2\pi)^{k/2} L_{\max} \frac{\sqrt{\det(\mathbf{Cov}_p)}}{\prod_{i=1}^k \Delta p_i} \quad (2.5)$$

where $\mathbf{Cov}_p$ is the covariance matrix of the fitted parameters and Δp_i are the prior parameter ranges. This is the interval over which each parameter was allowed to vary before fitting. $L_{\max}$ is the maximum likelihood at the best-fit parameters [\[Gregory, 2005, Haddad et al., 2024\]](#), given by:

$$L_{\max} = (2\pi s^2)^{-n/2} \exp\left(-\frac{1}{2s^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right) \quad (2.6)$$

which connects the BFI directly to the least-squares residuals. Here s^2 is the noise variance, replacing σ^2 . The BFI used by [Haddad et al. \[2024\]](#) assumes a known noise quantity. For this project, the true noise level is not known and is instead estimated from the fit residuals using the unbiased least-squares estimator [\[Montgomery et al., 2012\]](#):

$$s^2 = \frac{1}{n - k} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2.7)$$

Its use as a plug-in replacement for σ^2 in the BFI formula is expected to be

accurate when the number of data points greatly exceeds the number of parameters, as is the case here. This estimated variance is used in both the likelihood and covariance calculations. The covariance matrix is computed as:

$$\mathbf{Cov}_p = s^2 (\mathbf{A}^T \mathbf{A})^{-1} \quad (2.8)$$

The diagonal entries of $\mathbf{Cov}_p$ give the variance of each parameter (which is the uncertainty when square rooted), whereas the off-diagonal entries are the correlations (covariance) between them. The ratio $\sqrt{\det(\mathbf{Cov}_p)} / \prod \Delta p_i$ is the Occam factor. Where $\sqrt{\det(\mathbf{Cov}_p)}$ is the posterior volume, and $\prod \Delta p_i$ is the prior volume. This ratio measures what fraction of the allowed parameter space actually fits the data well. If the ratio is 1/100, it means only 1% of the prior range leads to a good fit. The Occam factor is thus asking the following question: “how much does the data actually narrow things down?”

The Occam factor penalises models if parameters are poorly constrained by the data (large variances), if parameters are strongly correlated (large covariances), and if prior ranges are too wide relative to the fitted parameter uncertainties [Dunstan et al., 2022, Haddad et al., 2024, MacKay, 1992]. The last point is important. If the prior volume is too large, but the posterior volume is small, this means the data has restricted the parameters to a small region within a large allowed range. While this alone is not problematic, the BFI penalises the model for exploring such a large parameter space. This penalty basically implies that there were other possible models, making the chosen optimum model less impressive.

If a model tries to introduce unnecessary parameters, it will see its improved likelihood offset by a reduced Occam factor, automatically implementing a preference for the simplest adequate model [Haddad et al., 2024, MacKay, 1992]. Strong correlations between parameters reduce the posterior volume, which is reflected by a decrease in $\det(\mathbf{Cov}_p)$. Thus, an improved fit does not necessarily mean that the model will be preferred.

Although Equation 2.8 assumes ordinary least squares, the non-negativity constraints in our fit do not affect the calculation as long as no coefficient sits at the boundary (at exactly zero). At the optimal model, all selected biomolecule coefficients are strictly positive, thus the covariance formula remains valid.

For practical comparison, the natural logarithm of the BFI is used. It is more convenient for computation, and the BFI is broken down into three interpretable

terms [Dunstan et al., 2022, Haddad et al., 2024]:

$$\begin{aligned} \ln(\text{BFI}) = & \underbrace{-\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(s^2) - \frac{n-k}{2}}_{\ln(L_{\max})} \\ & + \underbrace{\frac{k}{2} \ln(2\pi) + \frac{1}{2} \ln |\det(\mathbf{Cov}_p)|}_{\ln(\Omega)} - \underbrace{\sum_{i=1}^k \ln(\Delta p_i)}_{\text{prior term}} \end{aligned} \quad (2.9)$$

where $\ln(L_{\max})$ represents the likelihood, $\ln(\Omega)$ represents the Occam factor and the prior term penalises prior ranges. Note that the second and third terms are both the Occam factor, separated for better interpretation of penalty.

3 Data and Methodology

3.1 Data and Preprocessing

The Raman spectroscopic data used in this project was provided by [Gong et al. \[2025\]](#) at Queen Mary University of London. The dataset consists of a population-average Raman spectrum of *E. coli* in exponential (log) phase, where cells have high metabolic activity and divide rapidly, and stationary phase, where growth slows as nutrients are exhausted [[Wolfe, 2005](#)]. The reference spectra are a separate dataset with 15 spectra of individually purified biomolecules. All three datasets consist of 637 data points over the region 600.24 to 1750.45 cm^{-1} , known as the fingerprint region. The spectra of both growth phases are shown in Figure 3.1.

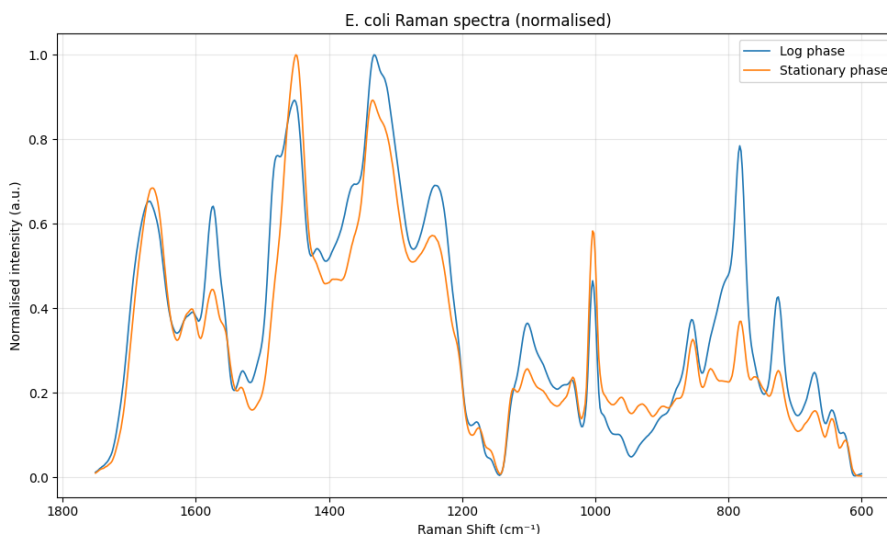


Figure 3.1: Normalised Raman spectra of *E. coli* in exponential (log) and stationary growth phases. Visible differences in peak intensities between the two phases reflect changes in biomolecular composition.

The 15 biomolecules in the reference library dataset are protein, RNA, glutathione (GSH), DNA, ATP, aspartate, glutamine, UTP, GTP, glucose, valine, fructose-1,6-bisphosphate, glutamate, lipid, and NAD^+ . They were selected based on their known abundance in *E. coli* cells [[Bennett et al., 2009](#), [Gong et al., 2025](#)]. Measured

under the same Raman conditions as the cellular spectra. All spectra were max-normalised to the range $[0, 1]$. Figure 3.2 shows the full reference library. Many biomolecules exhibit peaks in similar spectral regions, showing the degree of overlap and correlation between components.

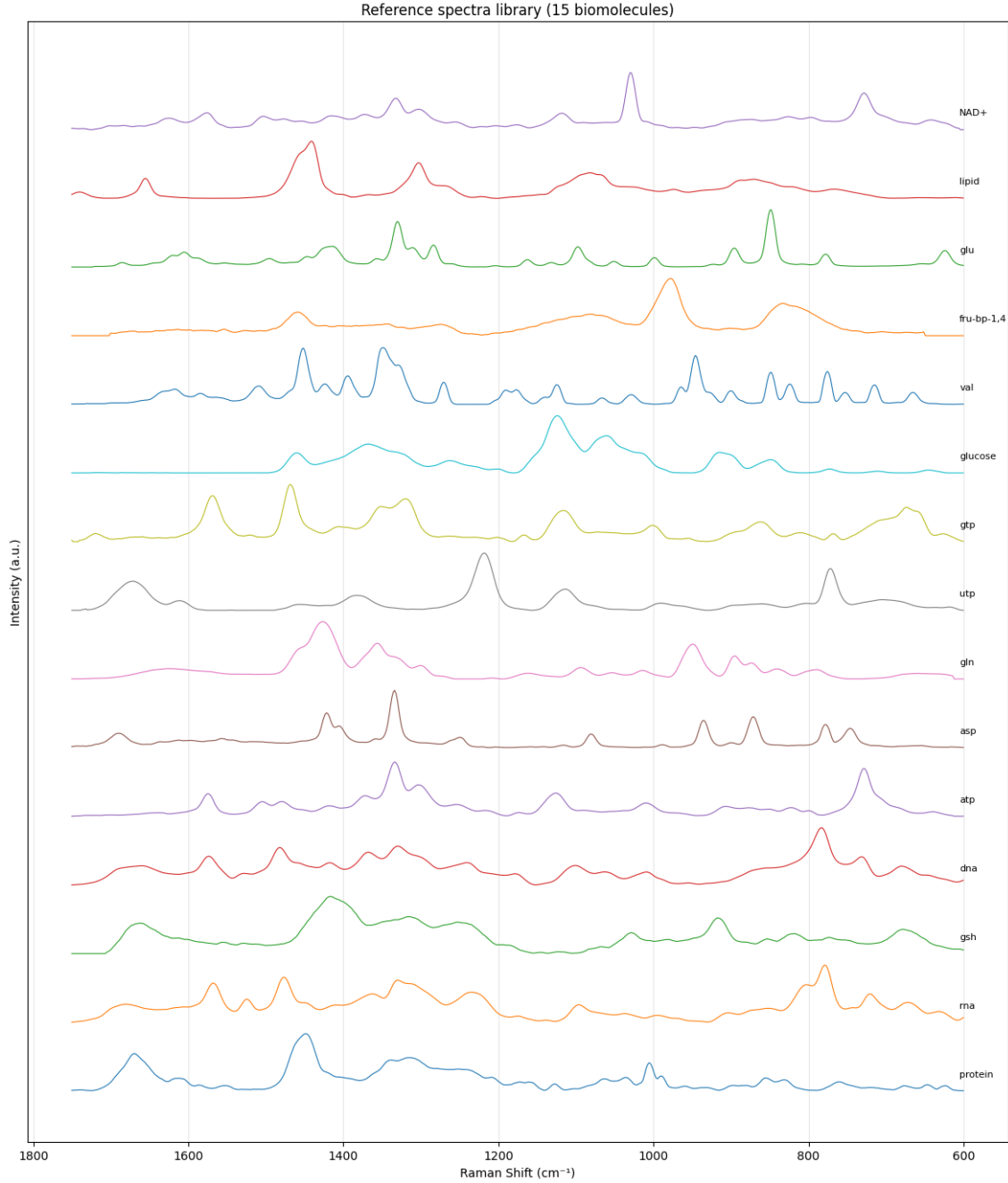


Figure 3.2: Reference Raman spectra of the 15 individually purified biomolecules used in the linear combination fitting, shown as a waterfall plot with each spectrum vertically offset for clarity. All spectra cover the region $600\text{--}1750\text{ cm}^{-1}$ and are max-normalised to $[0, 1]$.

The *E. coli* cells were grown in batch culture as described by [Gong et al. \[2025\]](#):

wildtype K12 strain BW25113, cultivated at 37°C in LB medium for 3 generations (exponential phase) or 7 generations (stationary phase). Raman microspectroscopy was performed using a Renishaw inVia confocal Raman microscope equipped with a 633 nm HeNe laser, with each spectrum accumulated from 20 iterations of 10-second exposure. The population-average datasets were obtained by averaging across 342 individual cells in log phase and 300 individual cells in stationary phase.

The only preprocessing step taken was applying max-normalisation to both spectra by dividing them by their maximum intensity value, placing them on the same $[0, 1]$ scale as the reference spectra. The residual background contributions were incorporated directly into the model through polynomial baseline terms. This differs from the approach of [Gong et al. \[2025\]](#), who applied smoothing and iterative multi-line background subtraction as preprocessing steps. We treat the baseline within the fit to avoid extra preprocessing assumptions and so that the baseline terms are included consistently in the fitted model.

The entire analysis pipeline was done in Python using Jupyter notebooks with the following libraries: Pandas used for data loading and handling [\[McKinney, 2010\]](#), NumPy for numerical computation [\[Harris et al., 2020\]](#), SciPy for least-squares fitting [\[Virtanen et al., 2020\]](#), and Matplotlib for plotting and visualisation [\[Hunter, 2007\]](#).

3.2 Model Fitting and Selection Procedure

We use the general linear combination model introduced in Section 2.1, where each model consists of a quadratic baseline and a subset of the reference spectra. The fitted spectrum was written as

$$y(\nu) = b_0 + b_1\tilde{\nu} + b_2\tilde{\nu}^2 + \sum_{i=1}^N c_i R_i(\nu) + \epsilon(\nu) \quad (3.1)$$

where $\tilde{\nu}$ is the wavenumber rescaled to the range $[-1, 1]$ so that it is centered around zero, b_0 , b_1 , and b_2 are the baseline coefficients, $R_i(\nu)$ are the biomolecular reference spectra included in the model, c_i are their fitted coefficients, and $\epsilon(\nu)$ is the noise term.

The model in Equation 3.1 is written in matrix form using Equation 2.2, where

the matrix $\mathbf{A}$ is constructed using the baseline terms and the reference spectra:

$$\mathbf{y} = \begin{bmatrix} 1 & \tilde{\nu}_1 & \tilde{\nu}_1^2 & R_1(\nu_1) & R_2(\nu_1) & \cdots & R_N(\nu_1) \\ 1 & \tilde{\nu}_2 & \tilde{\nu}_2^2 & R_1(\nu_2) & R_2(\nu_2) & \cdots & R_N(\nu_2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \tilde{\nu}_n & \tilde{\nu}_n^2 & R_1(\nu_n) & R_2(\nu_n) & \cdots & R_N(\nu_n) \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} + \boldsymbol{\epsilon} \quad (3.2)$$

The first three columns contain the baseline terms $(1, \tilde{\nu}, \tilde{\nu}^2)$, and the remaining columns contain the reference spectra included in the model (also seen in Table 3.1). A quadratic baseline was chosen as it is able to capture slow background variations without introducing unnecessary flexibility. A higher-degree baseline risks absorbing genuine spectral features. For a model containing N biomolecules, the matrix $\mathbf{A}$ has dimensions $637 \times (3 + N)$.

Table 3.1 below shows some rows of the matrix $\mathbf{A}$ for an example model containing two biomolecular reference spectra. Each row corresponds to one wavenumber point. The columns contain the baseline basis functions and the values for each biomolecule from the reference spectra at that wavenumber. In the full model, additional columns are added for the other selected biomolecular components.

$\nu \text{ (cm}^{-1}\text{)}$	1	$\tilde{\nu}$	$\tilde{\nu}^2$	protein	DNA
600.2412	1.0000	-1.0000	1.0000	0.0014	0.0811
602.2129	1.0000	-0.9966	0.9932	0.0026	0.0698
604.1836	1.0000	-0.9931	0.9863	0.0043	0.0599
606.1533	1.0000	-0.9897	0.9795	0.0039	0.0529
608.1230	1.0000	-0.9863	0.9728	0.0065	0.0498

Table 3.1: Example rows of the matrix $\mathbf{A}$ for a model containing a quadratic baseline and two example biomolecular reference spectra (protein and DNA). Each row corresponds to one wavenumber point. The first three columns give the baseline terms $(1, \tilde{\nu}, \tilde{\nu}^2)$, with the remaining columns containing the wavenumber values of the reference spectra.

The fit returns the set of coefficients that minimises the residual sum of squares between the measured and predicted spectra (as seen in Section 2.1). The residuals were used to estimate the noise variance s^2 (Equation 2.7) and to compute the

parameter covariance matrix (Equation 2.8).

The BFI (Equation 2.5) was evaluated in log form using Equation 2.9. The prior parameter ranges were chosen to match the ranges used in the fitting procedure, with the baseline coefficients allowed to vary over $[-0.5, 0.5]$ giving a prior width of $\Delta p_i = 1.0$. The biomolecular coefficients were constrained to $[0, 1]$, also giving a prior width of $\Delta p_i = 1.0$. This range is consistent with the requirement that reference spectra coefficients remain non-negative. For the baseline coefficients, $[-0.5, 0.5]$ is an appropriate range for normalised data. It is large enough that the baseline isn't artificially constrained and small enough that it can't overwhelm the biomolecular signal.

The fitting procedure starts from a baseline-only model. At each step, every reference spectrum is added in turn. Then the BFI is evaluated, and the spectrum giving the largest gain in $\ln(\text{BFI})$ is retained. The procedure repeats until all 15 reference spectra are exhausted. This produces a curve of $\ln(\text{BFI})$ as a function of the number of biomolecular components. The optimal model complexity is identified as the peak of this curve.

3.3 Comparison with Conventional Model Selection Criteria

To compare the BFI-guided results with conventional figures of merit, the same forward selection procedure was repeated separately for chi-squared (χ^2), reduced chi-squared (χ_ν^2), R^2 , AIC, and BIC. For R^2 , the model giving the largest value of the criterion was retained. For χ^2 , χ_ν^2 , AIC, and BIC it was the model giving the smallest value. The model scores for each were then tracked as a function of the number of biomolecular components as done for the BFI, allowing both the selection order and the preferred model complexity to be compared across criteria.

4 Results

4.1 Forward Selection and Optimal Model Complexity

We first begin with the BFI-guided forward selection procedure to determine the model complexity best supported by the data. Figures 4.1 and 4.2 show the $\ln(\text{BFI})$ score with the number of biomolecular components for the log and stationary phase spectra, respectively. In both cases, $\ln(\text{BFI})$ reaches a broad but identifiable maximum, corresponding to 8 biomolecular components for the log phase and 10 for the stationary phase.

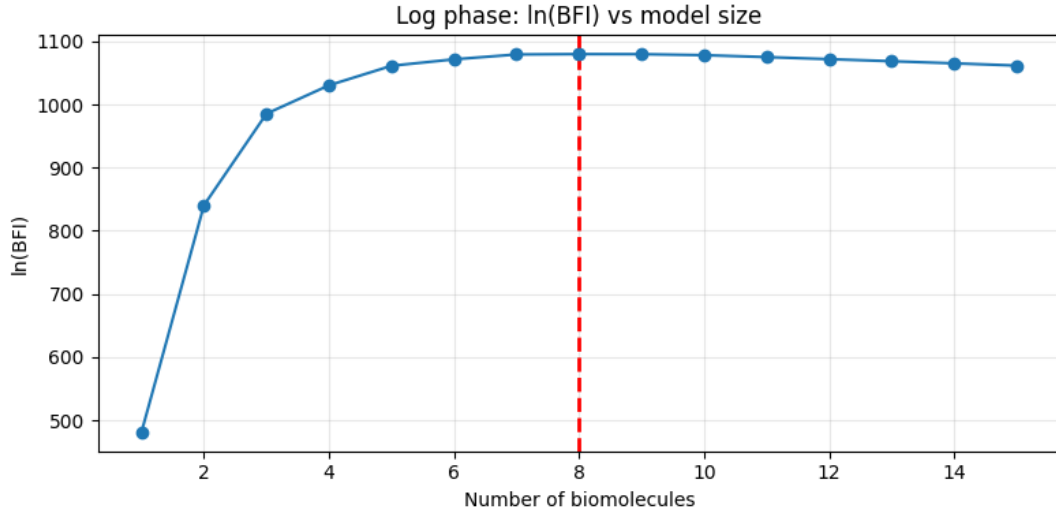


Figure 4.1: $\ln(\text{BFI})$ as a function of the number of biomolecular components included in the model for *E. coli* in exponential (log) growth phase.

The first few biomolecules improve the likelihood rapidly, overcoming their cost easily. As the amount of biomolecules N added nears the peak, the likelihood improvement becomes smaller while the Occam cost keeps accumulating. Complexity increases further until $\ln(\text{BFI})$ peaks and inevitably begins declining.

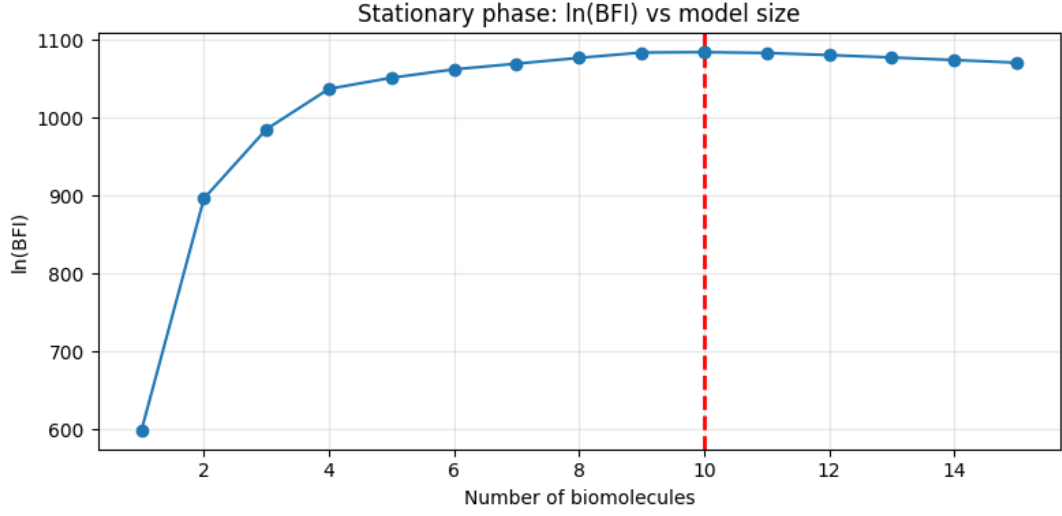


Figure 4.2: $\ln(\text{BFI})$ as a function of the number of biomolecular components included in the model for *E. coli* in stationary growth phase.

Figure 4.3 below shows the biomolecular components selected in the BFI-optimal models for both growth phases, along with their fitted coefficients. The phase differences are consistent with the fitting results reported by [Gong et al. \[2025\]](#), who also identified strong contributions from protein, RNA, DNA, glutathione, adenine nucleotides, and aspartate. [Gong et al. \[2025\]](#) used a background-noise threshold to identify significant biomolecular contributions.

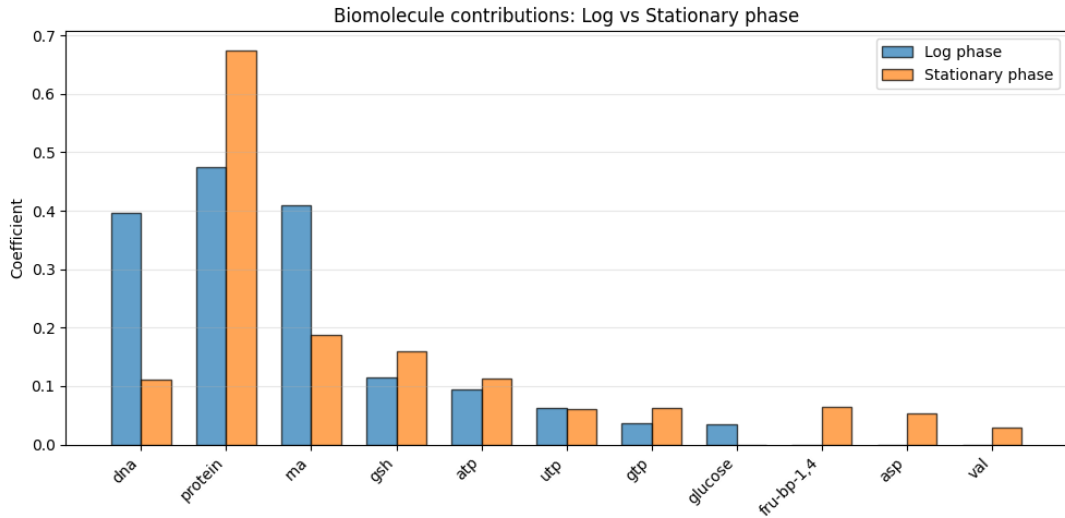


Figure 4.3: Comparison of the fitted biomolecule coefficients in the BFI-optimal models for the exponential (log) and stationary phase spectra of *E. coli*. Differences in coefficient value and selections highlight changes in the biomolecular composition supported by the data.

4.2 Fit Quality

Figures 4.4 and 4.5 below show the best-fit for both growth phases.

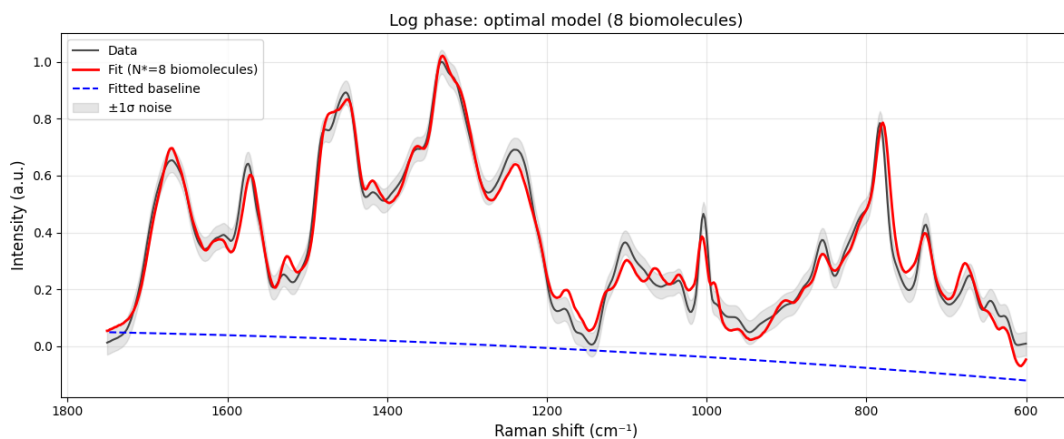


Figure 4.4: BFI-optimal fitted Raman spectrum for *E. coli* in exponential (log) growth phase.

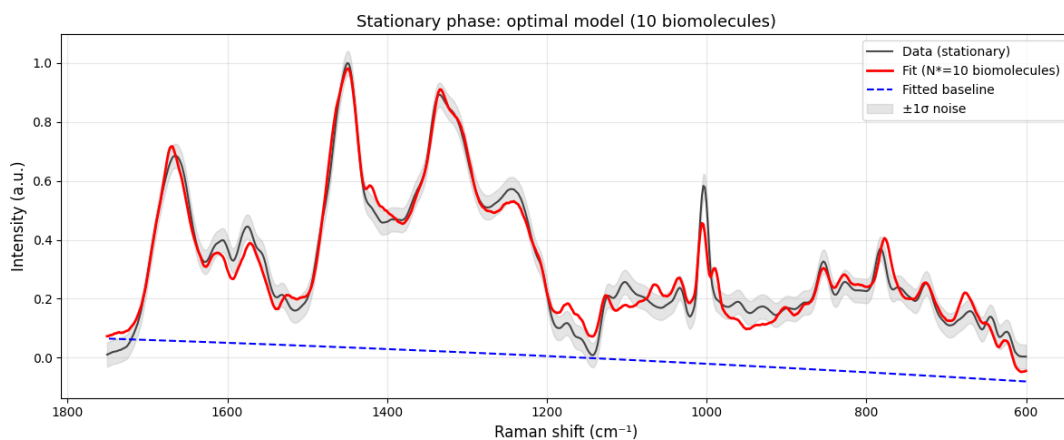


Figure 4.5: BFI-optimal fitted Raman spectrum for *E. coli* in stationary growth phase.

In both figures above, the fitted spectra reproduce the overall spectral features of the data well across the fingerprint region. The agreement is strongest in the broad high-intensity regions, while small deviations remain around sharper peaks and low intensity regions. The baselines are smooth, confirming that they do not absorb significant signal, and that the fit is mainly being explained by the reference spectra.

Figures 4.4 and 4.5 show that the selected biomolecules from the reference spectra are enough to capture dominant spectral contributions in each growth phase.

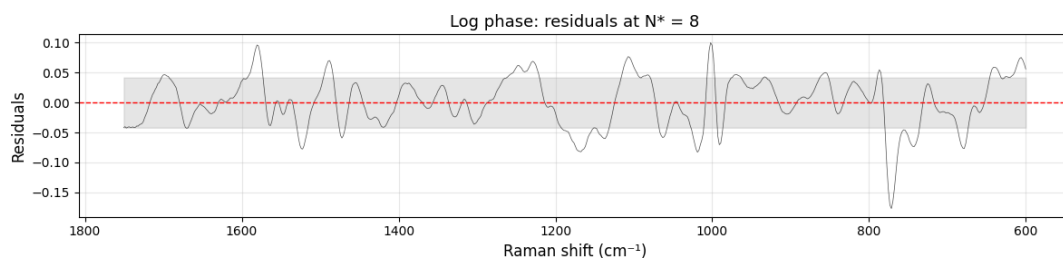


Figure 4.6: Residuals from the BFI-optimal fit for *E. coli* in exponential (log) growth phase.

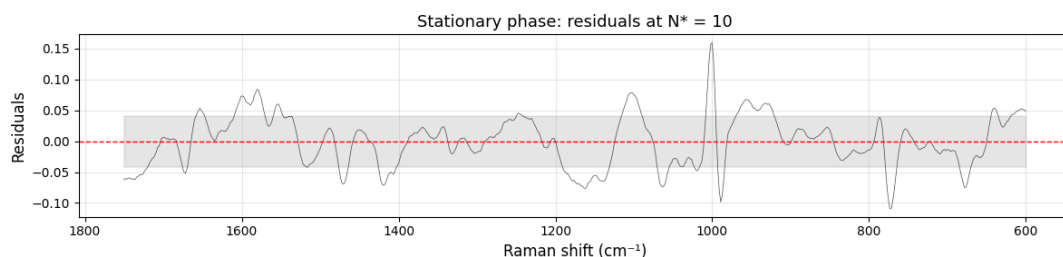


Figure 4.7: Residuals from the BFI-optimal fit for *E. coli* in stationary growth phase.

The residuals are the difference between the measured and fitted Raman spectrum, showing which parts of the data are not fully captured by the model. Figures 4.6 and 4.7 show the residuals for the log and stationary phase optimal models, respectively. The grey shaded region represents the estimated uncertainty around the fit. Residuals lying mostly within or close to this region indicate that the mismatch is small relative to the expected uncertainty of the model.

Figure 4.6 shows that the residuals for the log phase model are mostly within the expected uncertainty. However, the residuals are not completely random, with some recognisable structure particularly near 1500 cm^{-1} . Overlapping biomolecular contributions are the likely cause, or differences between the individually purified reference spectra and the same molecules within a cell.

Figure 4.7 shows similar behaviour, with residuals mostly within the expected range. There is some pattern at 1000 cm^{-1} , showing that the model is not fully capturing the spectral features in this region. This may be the selected 10-component model prioritising overall fit quality and complexity penalty rather than perfectly matching every peak.

4.3 Comparison of Model Selection Criteria

To assess how much the optimum model complexity depends on the choice of figure of merit, the forward selection procedure was repeated for the conventional criteria

introduced in Section 2.2. Table 4.1 summarises the optimal model complexity selected by each criterion for the two growth phases. In both phases, the BFI selected the smallest supported models, while AIC, χ^2 , reduced χ^2 , and R^2 generally favoured slightly larger models. BIC gave the closest agreement with the BFI, differing by only one component in the log phase and selecting the same model size in the stationary phase.

Table 4.1: Optimal number of biomolecular components selected by each model-selection criterion for exponential (log) and stationary phase spectra.

Criterion	Log phase N^*	Stationary phase N^*
ln(BFI)	8	10
χ^2	10	12
Reduced χ^2	10	12
AIC	10	11
BIC	9	10
R^2	10	12

In all cases, the first selected biomolecules were the same or very similar, with the larger models differing only through the addition of weak extra contributors. The main spectral structure is already captured by the BFI-selected models, while the later-stopping criteria mainly extend the model with components that make unnecessary contributions. It is worth noting that R^2 , χ^2 and reduced χ^2 had ambiguous stopping points, with a plateau instead of a peak. For the stationary-phase spectrum, the BFI and BIC resulting models contained the same set of biomolecular components.

To determine whether the larger models produce a meaningfully improved model, the stationary-phase BFI-optimal fit was compared with the fit obtained using the R^2 criterion. The R^2 criterion was chosen because it selected a larger model, with 12 biomolecular components instead of 10. The additional components are assessed to see whether they lead to a meaningful improvement in the fit. Figure 4.8 below shows the fitted stationary-phase spectrum obtained using the R^2 -selected model, and Figure 4.9 shows its residuals.

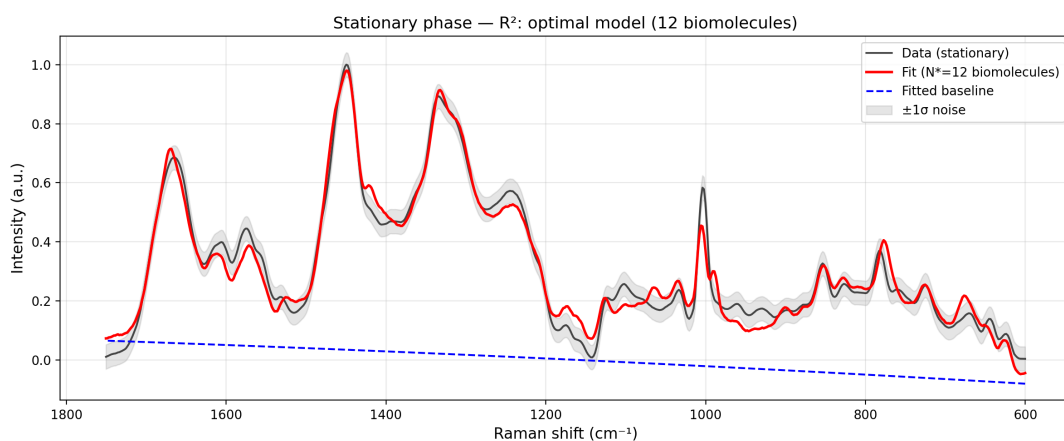


Figure 4.8: R^2 -selected fitted Raman spectrum for *E. coli* in stationary growth phase.

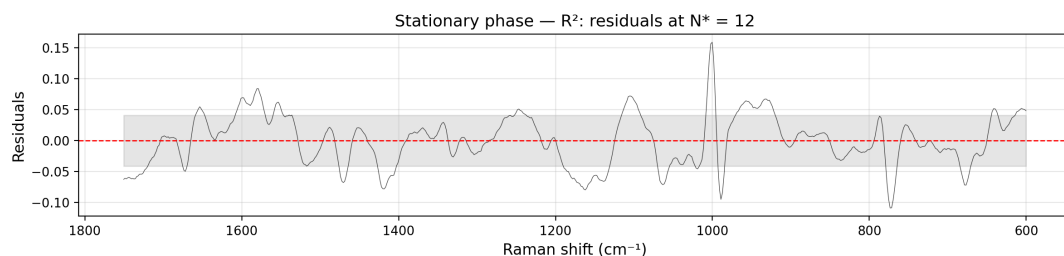


Figure 4.9: Residuals from the R^2 -selected fit for *E. coli* in stationary growth phase.

Compared with the BFI-optimal fit in Figure 4.5, the R^2 -selected fit in Figure 4.8 is visually very similar, with no substantial visible change. The additional components thus do not substantially improve the model, introducing unnecessary additional complexity. The fitted spectrum is slightly improved, with the change being small and the residuals in Figure 4.9 remaining very similar to those of the BFI-optimal model. This suggests that the extra components are primarily being used to absorb weak overlapping structure and noise, signalling overfitting.

5 Discussion

5.1 Implications of Main Findings

The results show that the measured Raman spectra can be described using a limited subset of the reference biomolecules. While the different selection criteria identified very similar dominant components, they began to disagree in the final stages of selection, where the remaining candidates made relatively small improvements.

The main uncertainty in this entire approach is whether weak additional components provide genuinely meaningful information or simply produce slight improvements to the fit. In this situation, a penalty for parameter correlation is especially important, since a small improvement in fit quality does not necessarily mean that the extra biomolecular components are truly supported by the data. This helps explain why the BFI selected smaller models than most conventional criteria, and hence why the additional components selected by other criteria should be interpreted cautiously, as it does not accept minor improvements in residuals or components that are correlated with others already in the fit. What was interesting about the comparison with other criteria was how close BIC was to the BFI in terms of model complexity. This is likely because BIC penalises complexity more strongly than the other figures of merit.

5.2 Limitations of the approach

The forward selection approach is a ‘greedy’ algorithm, as it adds the biomolecular component that gives the largest improvement at each step. This means that it does not guarantee the globally optimal model, although the resulting model is expected to be near-optimal. Of course, if every possible combination of the reference spectra were tested and compared using the BFI, a similar or better model could be found.

Another limitation is that the noise variance is estimated from the fit residuals (Equation 2.7) rather than measured independently. This is a reasonable estimate, but it is not an exact treatment of all uncertainty. Using an independently measured

noise term would be preferred and more accurate.

The reference library itself is also restricted in scope. The model is restricted to the available reference spectra, while the Raman data may contain contributions from biomolecular components not present in that library. Therefore, unless the reference library is known to contain all relevant contributors, the fitted model should not be interpreted as a complete biochemical description of the cell.

Finally, for this project the prior term does not have a penalty effect in the BFI because the prior ranges were equal to 1, which makes the log form term equal to zero. This is not an issue for this project due to the Occam factor penalty providing the main penalty of correlation and complexity, but it is worth taking into account for future work where prior widths will be different.

6 Conclusion

This project aimed to determine the true biomolecular information content of Raman spectroscopic data for *E. coli*. The results showed that 8 out of the 15 biomolecules from the reference library were optimal for describing the exponential (log) phase spectrum, while 10 reference biomolecules were optimal for describing the stationary phase spectrum. The selected biomolecules were able to reproduce the spectra well, whilst conventional criteria generally selected more complex models that produced no visible improvement in fit quality. The BFI therefore provides a useful stopping rule that takes parameter correlation into account and avoids unnecessary model complexity. Future work could extend this approach to larger biomolecular reference libraries and spectroscopic datasets, where spectral overlap and biological complexity increase. It would also be worth testing the BFI on data with an independent noise term readily available to assess the difference between using estimated and measured noise.

Bibliography

- H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. doi: 10.1109/TAC.1974.1100705.
- B. D. Bennett, E. H. Kimball, M. Gao, R. Osterhout, S. J. Van Dien, and J. D. Rabinowitz. Absolute metabolite concentrations and implied enzyme active site occupancy in *Escherichia coli*. *Nature Chemical Biology*, 5(8):593–599, 2009. doi: 10.1038/nchembio.186.
- H. J. Butler, L. Ashton, B. Bird, G. Cinque, K. Curtis, J. Dorney, K. Esmonde-White, N. J. Fullwood, B. Gardner, P. L. Martin-Hirsch, M. J. Walsh, M. R. McAinsh, N. Stone, and F. L. Martin. Using Raman spectroscopy to characterize biological materials. *Nature Protocols*, 11(4):664–687, 2016. doi: 10.1038/nprot.2016.036.
- D. J. Dunstan, J. Crowne, and A. J. Drew. Easy computation of the Bayes factor to fully quantify Occam’s razor in least-squares fitting and to guide actions. *Scientific Reports*, 12:993, 2022. doi: 10.1038/s41598-021-04694-7.
- Z. Gong, A. Sapelkin, and C. Engl. Improved data analysis for non-invasive, label-free biomolecular characterisation of single bacterial cells via Raman microspectroscopy. *Research Square*, 2025. doi: 10.21203/rs.3.rs-6673956/v1. Preprint.
- P. C. Gregory. *Bayesian Logical Data Analysis for the Physical Sciences: A Comparative Approach with Mathematica Support*. Cambridge University Press, 2005.
- L. Haddad, D. Gianolio, D. J. Dunstan, Y. Liu, C. Rankine, and A. Sapelkin. Quantifying intuition: Bayesian approach to figures of merit in EXAFS analysis of magic size clusters. *Nanoscale*, 16:5768–5775, 2024. doi: 10.1039/D3NR05110B.
- C. R. Harris et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- J. D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/MCSE.2007.55.
- K. S. Lee, Z. Landry, F. C. Pereira, M. Wagner, D. Berry, W. E. Huang, G. T. Taylor, J. Kneipp, J. Popp, M. Zhang, J.-X. Cheng, and R. Stocker. Raman microspectroscopy for microbiology. *Nature Reviews Methods Primers*, 1:80, 2021. doi: 10.1038/s43586-021-00075-6.
- D. J. C. MacKay. Bayesian interpolation. *Neural Computation*, 4(3):415–447, 1992. doi: 10.1162/neco.1992.4.3.415.

- W. McKinney. Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010. doi: 10.25080/Majora-92bf1922-00a.
- D. C. Montgomery, E. A. Peck, and G. G. Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, Hoboken, NJ, 5 edition, 2012. ISBN 978-0-470-54281-1.
- C. V. Raman and K. S. Krishnan. A new type of secondary radiation. *Nature*, 121: 501–502, 1928. doi: 10.1038/121501c0.
- G. Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2): 461–464, 1978. doi: 10.1214/aos/1176344136.
- P. Virtanen et al. Scipy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- A. J. Wolfe. The acetate switch. *Microbiology and Molecular Biology Reviews*, 69(1):12–50, 2005. doi: 10.1128/MMBR.69.1.12-50.2005.